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Is zero more than nothing? Relations between concepts of zero and integer understanding

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ABSTRACT

The concept of zero is challenging for many children. This study investigated conceptions of zero in fifth- through seventh-grade children ($N = 72$) in the United States and examined how children's conceptions of zero related to other aspects of their integer knowledge. At the outset of the study, many participants held a null conception of zero (i.e., zero as “nothing”), and some participants held a conception of zero as the symmetry point between the positive and negative integers. Participants in higher grades were more likely to hold a symmetry conception. We hypothesized that participants' conceptions of zero would be related to other aspects of their integer knowledge. Relative to participants who held a null conception of zero, participants who held a symmetry conception demonstrated greater knowledge of the additive inverse principle (for every number x , there exists an inverse, $-x$, such that the two numbers sum to zero) and greater integer arithmetic skill. Conceptions of zero were not related to integer magnitude understanding. We also examined whether a brief lesson focusing on zero as the symmetry point would lead to shifts in participants' conceptions of zero and gains in understanding of the additive inverse principle. Participants were randomly assigned to receive a lesson about zero as null or zero as the symmetry point between positive and negative integers. Relative to the lesson about zero as null, the lesson about zero as the symmetry point did not lead to substantial changes in conceptions of zero or additive inverse knowledge.

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Introduction

Early numerical development involves understanding fundamental concepts and procedures for natural numbers {1, 2, 3 ...}, including one-to-one correspondence and cardinality (Gelman & Gallistel, 1986; Sarnecka & Carey, 2008). Researchers in mathematical cognition and mathematics education have posited that natural number concepts serve as a foundation for later-acquired mathematical concepts such as integer concepts (e.g., Bofferding, 2014; Varma et al., 2019). Integers extend the natural numbers with zero and negative whole numbers.

Although many preschool children demonstrate understanding of counting and basic natural number concepts, the concept of zero remains challenging for many early elementary school children (Bialystok & Codd, 2000; Hartmann et al., 2022; Krajcsi et al., 2021; Wynn, 1998). Children's early understanding of zero often involves misconceptions such as "zero is not a number" or vague ideas such as "zero is nothing" (Lappan & Wheeler, 1987). Not until 5 or 6 years of age do most children understand that zero is a number and correctly identify it as the smallest natural number (Wellman & Miller, 1986).

Negative numbers also pose many challenges for learners. One source of the challenge may be that, unlike natural numbers, negatives do not have an obvious mapping to physical objects or perceptual representations. For example, "one" and "two" readily map onto quantities of physical objects (e.g., one jellybean, two jellybeans). It is much more difficult to display negative one or negative two jellybeans.

Many scholars have considered the perceptual, motor, and experiential bases of number concepts (e.g., Fischer & Shaki, 2018; Newcombe et al., 2015; Piaget, 1965). One general account draws on *conceptual metaphor theory*, which holds that people understand abstract concepts by mapping them to familiar source domains such as bodily experiences (e.g., Lakoff & Johnson, 1980). Applying this general theory to number, Lakoff and Núñez (2000) proposed that people conceptualize number via a small set of grounding metaphors, which allow people to project from everyday experiences to abstract concepts. Two of the primary metaphors Lakoff and Núñez discussed are *numbers as collections of objects* and *numbers as locations along a path*. Each of these metaphors provides a way of conceptualizing mathematical actions and relations. For example, within the metaphor of *numbers as collections of objects*, arithmetic is conceptualized in terms of adding to or taking away from a collection, and within the metaphor of *numbers as locations along a path*, arithmetic is conceptualized in terms of moving in a specific direction along a path.

Lakoff and Núñez argued that the *numbers as a collections of objects* metaphor is learned early in development, based on everyday experiences of engaging in actions with small sets of objects and subitizing those sets. Given these experiences, the natural numbers are easily mapped to collections of objects. Lakoff and Núñez (2000) argued that the metaphor of *numbers as locations along a path* is a later construction that relies on recognizing relations between a path of motion and a physical segment, with the origin of the movement corresponding to one end of the segment and the end of the movement corresponding to the other end. The natural numbers can also be readily mapped to locations along a path, for example, the locations one traverses while taking a series of steps.

Given children's early facility with perceptual and object-based representations of natural numbers and the early-developing, grounded conceptual metaphor of *numbers as collections of objects*, it seems likely that children's earliest understanding of zero reflects the absence of objects, or the empty set. We refer to this conception as the *null* conception of zero because it reflects the idea that zero represents nothingness, or the absence of quantity, akin to the null set in mathematics.

Negative numbers present a challenge for the *numbers as collections of objects* metaphor because they do not readily lend themselves to perceptual or object-based representations. However, it is straightforward to conceive of negatives as *locations along a path*. Indeed, Lakoff and Núñez (2000) noted that the *locations along a path* metaphor provides a ready mapping to zero, as the origin of the path, and a "natural extension to negative numbers" as "point-locations on the other side of zero from the positive numbers along the same path" (p. 73). We suggest that the conceptual metaphor of *locations along a path* supports thinking about zero as the location between the positive and negative numbers, which we term a *symmetry* conception of zero.

The conceptual metaphor of numbers as locations along a path is aligned with theoretical and empirical work focusing on spatial representations of numerical concepts, which goes back as far as Galton (1880). More recent work has shown that, when making odd/even judgments, people in Western cultures respond more quickly to large numbers with the right hand and to small numbers with the left (e.g., Dehaene et al., 1993; Wood et al., 2008)—a phenomenon that implies the existence of a horizontal, left-to-right, spatial mental representation for numbers, termed the “mental number line.” Children display behavioral evidence for the mental number line for natural numbers (e.g., Aulet & Lourenco, 2018; Tam et al., 2019), and with experience and practice, people begin to represent zero and negative numbers on the mental number line as well (Sella et al., 2016; Young & Booth, 2015).

How do learners come to understand negative number magnitudes? Varma and Schwartz (2011) proposed a learning progression for integers that is compatible with the conceptual metaphor of *numbers as locations along a path*. They argued that before children learn integers, they represent natural number magnitudes in terms of a mental number line—which can be construed as an application of the locations along a path metaphor for natural numbers. Because this early mental number line is limited to natural numbers, children initially understand negative numbers in terms of symbolic rules such as “negative numbers are always less than positive numbers.” Thus, children’s early understanding of negative numbers is a rule-based conception that does not truly incorporate magnitude (see also Blair et al., 2013).

According to Varma and Schwartz (2011), with time and experience, children transform their representations of integers by incorporating properties unique to the integers, such as the *additive inverse* property, which is the principle that for every number there exists an inverse such that the two numbers sum to zero (i.e., $x + -x = 0$). This property involves symmetry-based reasoning that *reflects* the mental number line for natural numbers to the other side of zero to represent negative numbers, in what Varma and Schwartz (2011) termed a *reflection point* model. This reflection of the mental number line for natural numbers creates new locations along a path to the left of zero for the negative numbers, and zero is the symmetry point of this reflected number line. In the reflection point model, negative integers inherit magnitude from the natural numbers, so once children achieve a symmetry-based view, they represent the magnitudes of negative numbers.

Some empirical evidence supports the view that children progress from a rule-based conception of negative numbers to a symmetric, reflection-based conception. Varma and Schwartz (2011) asked sixth-grade children and adults to make magnitude comparisons (e.g., “Choose the greater number: 2, -4”) for different types of number pairs (e.g., positive–positive, positive–negative, negative–negative) and examined whether participants displayed the “distance effect” (Moyer & Landauer, 1967), which is the phenomenon that people judge pairs of numbers with larger distances (e.g., 3, 9) more quickly than pairs with smaller distances (e.g., 5, 3) and vice versa. If people display a distance effect, this implies that they have activated the magnitudes of the numbers in the pairs. Varma and Schwartz (2011) found that children did *not* display a distance effect for positive–negative pairs (e.g., 2, -4) but instead judged both near and far positive–negative pairs with similar reaction times (RTs). These data suggest that children did *not* activate the magnitudes of the negative numbers but instead used the “positives are greater than negatives” rule.

In contrast, adults’ reaction time patterns suggested that they used a reflection point model in comparing magnitudes. According to Varma and Schwartz (2011), the reflection point model should be manifested in an *inverse* distance effect on mixed comparisons (i.e., positive–negative comparisons)—that is, *slower* RTs for pairs that are farther apart. This pattern should arise due to the psychophysical scaling of magnitude representations of the mental number line for natural numbers and the corresponding reflection for negative numbers (Varma et al., 2019). Briefly, mixed pairs that are close together (e.g., -2, 1) correspond to magnitudes that are further apart in psychophysical terms (e.g., 2 vs. 1) and therefore are fast and easy to distinguish, whereas mixed pairs that are further apart (e.g., -5, 6) correspond to magnitudes that are closer together in psychophysical terms (e.g., 5 vs. 6), so they are slower and more difficult to distinguish, yielding an inverse distance effect (see Varma et al., 2019, for discussion). As predicted, adult participants showed an inverse distance effect (Varma & Schwartz, 2011), in contrast to the absence of a distance effect observed in children.

Although Varma and Schwartz (2011) did not focus on learners’ understanding of zero, their hypothesized progression has implications for concepts of zero. Specifically, this progression suggests

that as children's view of the number line shifts to incorporate the reflection of the positive numbers, children should begin to view zero as the point on the number line that divides the positives and the negatives. Thus, we hypothesize that learning about negative numbers as a reflection of positive numbers may lead to a shift in children's concepts of zero.

What sorts of experiences or instructional activities might foster movement along this learning progression? One possibility is practice with activities that focus on the reflection of the natural numbers. Tsang et al. (2015) developed an experimental curriculum that focused on reflection, specifically involving practice in *folding* number lines. The curriculum used a specially designed number line manipulative that could be folded in half. This curriculum helped fourth-grade students learn to solve complex problems involving negative numbers (i.e., negative fractions) more effectively than either of two different control curricula, one that involved jumping along a number line and one that involved a cancellation procedure.

In the current study, we took a different approach to encouraging learners to think about the idea of reflection. Specifically, we taught children about inverse number pairs (e.g., 3 and -3) and demonstrated on a number line that they are equidistant from zero. These ideas underpin the *additive inverse principle* (i.e., for every number, there exists an inverse such that the two numbers sum to zero). Placing inverse pairs on the number line highlights that x and $-x$ can be conceptualized as locations on the number line that are equidistant from zero and thereby highlights that zero is the midpoint between positives and negatives. We suggest that drawing attention to this equidistance may both support children's understanding of the magnitudes of negatives and help children to develop a conception of zero as the symmetry point. We compared children's progress with this lesson with children's progress with a lesson that focused on zero as the absence of objects, or the empty set—that is, a *zero-null* conception. We did not expect a lesson focusing on zero as the empty set to foster progress to a view of zero as the symmetry point between positives and negatives.

Research questions

In this study, we addressed four questions. First, what is the nature of children's concepts of zero in late elementary and middle school? We hypothesized that (1a) most participants will demonstrate a concept of zero as null, and some will demonstrate a concept of zero as the symmetry point between the negative and positive integers. We further hypothesized that (1b) the proportion of participants who display a symmetry conception will increase across grade levels.

Second, are children's concepts of zero associated with their understanding of integers? We hypothesized that (2a) participants who have a concept of zero as the symmetry point will have greater understanding of the additive inverse principle than participants who have a null concept and that (2b) participants who have a concept of zero as the symmetry point will be more fluent with integer arithmetic.

Third, will a brief instructional intervention about integers focusing on symmetry enhance children's concepts of zero and their understanding of the additive inverse principle? We hypothesized that, relative to an intervention that focuses on a null conception of zero, (3a) an intervention focusing on symmetry will support participants in developing an understanding of zero as the symmetry point and (3b) an intervention focusing on symmetry will support participants' understanding of the additive inverse principle.

Fourth, do children's initial conceptions of zero predict whether they activate the magnitudes of negative numbers when they compare positive and negative numbers? Based on past work, we expected that participants who represent magnitudes of negative numbers in terms of their distance from zero (i.e., as symmetrical to the positive numbers) should demonstrate a distance effect for positive-negative pairs (specifically, an inverse distance effect; Varma et al., 2019). In contrast, participants who represent all negative numbers as "less than zero"—that is, who do *not* automatically activate the distinct magnitudes of negative numbers—should *not* demonstrate a distance effect. Instead, they should judge all positive-negative pairs with similar speed. We expected that participants who hold a symmetry conception of zero will activate magnitudes for negative numbers; therefore, we hypothesized that (4a) participants who hold a symmetry conception of zero should demonstrate a distance effect for positive-negative pairs on the magnitude comparison task. In

contrast, participants who hold a null conception of zero should have less differentiated magnitude representations for negative numbers, so (4b) participants who hold a null conception of zero should be unlikely to demonstrate a distance effect for positive–negative pairs on the magnitude comparison task. Note that although we did expect our brief intervention to affect participants' concepts of zero, we did not expect immediate effects on implicit understanding of magnitudes, so we did not expect to find intervention effects on magnitude comparison.

We addressed these research questions in an experiment with a pretest–intervention–posttest design. The intervention included game-based tasks that highlight a concept of either zero as null or zero as the symmetry point between positive and negative integers.

Method

Participants

A total of 73 children participated in this study. One participant was excluded because of procedural issues during the study. Thus, the final sample included 72 children ($M_{\text{age}} = 11.43$ years, range = 10.17–13; 30 girls, 41 boys, and 1 nonbinary; 54 White, 6 Asian, 1 Hispanic or Latino, 5 African American, and 6 more than one race or ethnicity).

We initially preregistered that we would include participants in Grades 4 to 7; however, after working with several fourth-grade students, we realized that the experimental sessions were too lengthy for these young children, so we limited our data collection to participants in Grades 5 to 7. We selected this age range because children in the United States typically receive formal instruction about negative integers beginning in sixth grade (National Governors Association Center for Best Practices & Council of Chief State School, 2010). Of the 72 participants, 33 were in fifth grade, 25 were in sixth grade, and 14 were in seventh grade. We tested more children in lower grades because we aimed to test the effects of an intervention about the symmetry conception of zero, and we wanted to ensure that many participants did not already hold this conception at the outset of the study.

The sample was a convenience sample recruited from a mid-sized midwestern city in the United States. Participants were recruited from a list of families who had previously participated in studies in the research lab, and each participant was compensated \$15.

Stimuli and design

The study used a pretest–lesson–posttest design. Each child participated in one hour-long, one-on-one Zoom session moderated by an experimenter. See Fig. 1 for a schematic of the study design. Materials, stimuli, and the preregistration are available on the Open Science Framework (<https://osf.io/km3tz>).

Pretest

Participants completed a pretest to assess concepts of zero and understanding of the additive inverse principle (see Appendix A). The pretest included three items to assess concepts of zero: two direct questions (What is zero? and Is zero a number?; Wheeler & Feghali, 1983) and a number line placement item in which participants placed the number 3 in one of a set of pre-marked locations on a number line. The pretest also included six items to assess understanding of the additive inverse principle: three problem-solving items (e.g., Solve and explain: $6 + 5 + -5 = \underline{\quad}$), an item intended to elicit mention of the principle (e.g., Why is this problem false? $7 + -8 = 0$), and two worked examples of problems that indicated use of the principle. In addition, participants completed an additive identity principle item (Why is this problem true? $3 + 0 = 3$).

Lesson

Next, participants engaged in a game-based lesson. Participants were randomly assigned to either the *zero–null* condition, in which they received a lesson focusing on zero as the empty set, or the *zero–symmetry* condition, in which they received a lesson focusing on zero as the midpoint between the



Fig. 1. Schematic of study procedure.

positive and negative numbers on the number line. Each lesson included two activities: a card-sorting game and a fill-in-the-blank task.

In both conditions, the card-sorting game required participants to find four pairs of cards from a set of 20 cards. Participants' activities with the cards depended on their condition assignment. In the card-sorting game in the zero-symmetry condition, participants identified pairs made up of a positive integer and the corresponding negative (i.e., an additive inverse pair) and located both values on a number line. Qualtrics displayed a labeled number line with 10 ticks to the left of zero and 10 ticks to the right of zero and the set of 20 cards that included green positive numbers and corresponding red negative numbers. The set of cards contained four green-red pairs of matching positive-negative numbers (e.g., green 4, red -4). The experimenter said, "You're going to see a set of 20 green and red numbers. We want you to match a positive green number and the corresponding red negative number. When you find a match, a number line will show. Please choose the numbers' respective places on the number line." After the participant located the pair, the experimenter emphasized, "Yes! Because x is x units away from zero and $-x$ is x units away from zero" (substituting the relevant number for x in each case), emphasizing the equidistance of the members of each pair from zero. Then, the participant was asked, "What is exactly between x and $-x$?", for which the only choice was zero. This procedure was repeated for four pairs (see Fig. 2).

In the card sorting game in the zero-null condition, participants identified pairs of positive integers (e.g., 4 and 4), and they moved the corresponding number of discs from one box to another, emptying the first box. Qualtrics displayed one box outlined in green and one box outlined in red as well as the set of cards that included both green and red numbers. The set of cards contained four green-red pairs of matching numbers (e.g., green 4 and red 4). When the participant chose a pair of numbers, the green box displayed the corresponding number of black discs (e.g., 4). The experimenter said, "You're going to see a set of 20 green and red numbers. I want you to match a green number and the corresponding red number. When you find a match, two boxes will show." After the participant chose a pair, the experimenter said, "Great. You found the pair of x 's. Remember, the goal of the game is to empty the green box so that it has zero discs in it. You can move the discs from the green box to the red box. How many discs do you need to move into the red box so that the green box is empty and has zero discs in it?" After the participant emptied the green box, the experimenter emphasized, "Yes! Look here, the green box is empty and has zero discs in it now." Then, the participant was asked, "How many discs were in the green box at the end of our activity?", for which the only choice was zero. This procedure was repeated for four pairs (see Fig. 3).

Next, participants in each condition performed a fill-in-the blank task in which they generated numbers to make a statement true. Each participant completed five statements, and the nature of those statements depended on participants' condition (see Appendix B).

In the zero-symmetry condition, the statements related to the idea that inverse number pairs are equidistant from zero (e.g., "Both $\underline{\quad}$ and $\underline{\quad}$ are 3 units away from 0"). When a participant correctly filled in the blanks with the appropriate numbers, the experimenter emphasized that the two numbers of the pair were equidistant from zero. For instance, the experimenter said, "Correct! Both 3 and -3 are 3 units away from 0 on the number line."

In the zero-null condition, the statements related to the idea of zero as the empty set (e.g., "A box has 9 discs. (a) $\underline{\quad}$ discs and then 3 additional discs are pulled out so that there are 0 discs left in the box and it is empty."). When a participant correctly filled in the blank with the appropriate number, the experimenter emphasized the significance of zero as the empty set. For instance, the experimenter

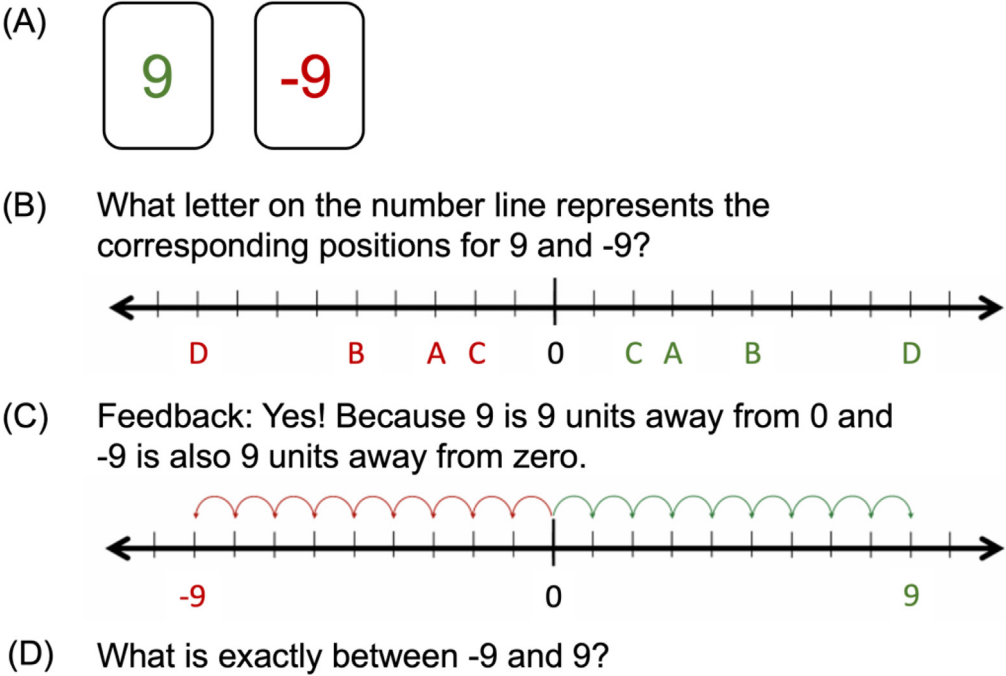


Fig. 2. Sample of card-sorting task used during intervention in the zero-symmetry condition.

said, “Correct! There are 9 discs in the box. If you pull out 6 discs and then 3 additional discs, then there will be 0 discs left and the box will be empty.”

Posttest

After the lesson, participants completed the posttest, which was isomorphic to the pretest.

Magnitude comparisons

Next, participants engaged in a timed magnitude comparison task adapted from [Varma and Schwartz \(2011\)](#). This task was also administered only after the posttest both because we did not wish to fatigue participants prior to the intervention and because we did not have specific hypotheses about intervention effects for this task.

In this task, participants were presented with number comparisons in PsychoPy ([Peirce et al., 2019](#)) hosted online via Pavlovia. The integers were grouped into five sets: small natural numbers {1, 2, 3, 4}, large natural numbers {6, 7, 8, 9}, small negative integers {−1, −2, −3, −4}, large negative integers {−6, −7, −8, −9}, and zero {0}. An integer was considered small if $|x| < 5$ and large if $|x| > 5$.

The stimuli were pairs of integers that formed two fully crossed within-participants factors: *comparison type* and *distance*. Comparison type had four levels: positive, negative, mixed, and zero. Positive comparisons involved comparing two positive numbers, negative comparisons involved two negative numbers, mixed comparisons involved one positive number and one negative number, and zero comparisons involved zero and either a positive or negative number. Distance had two levels: near and far. Distance was determined by the mathematical distance between the integers being compared ($|x - y|$). Near-distance stimuli had a distance of 2 or 3, and far-distance stimuli had a distance of 7 or 8. For positive, negative, and zero comparisons, near-distance stimuli were created by pairing two small integers or two large integers. Far-distance stimuli were created by pairing a small integer and a large integer. For mixed comparisons, near-distance stimuli paired a small positive integer and a small negative integer (−2 vs. 1), and far-distance stimuli paired a small negative integer and a large positive

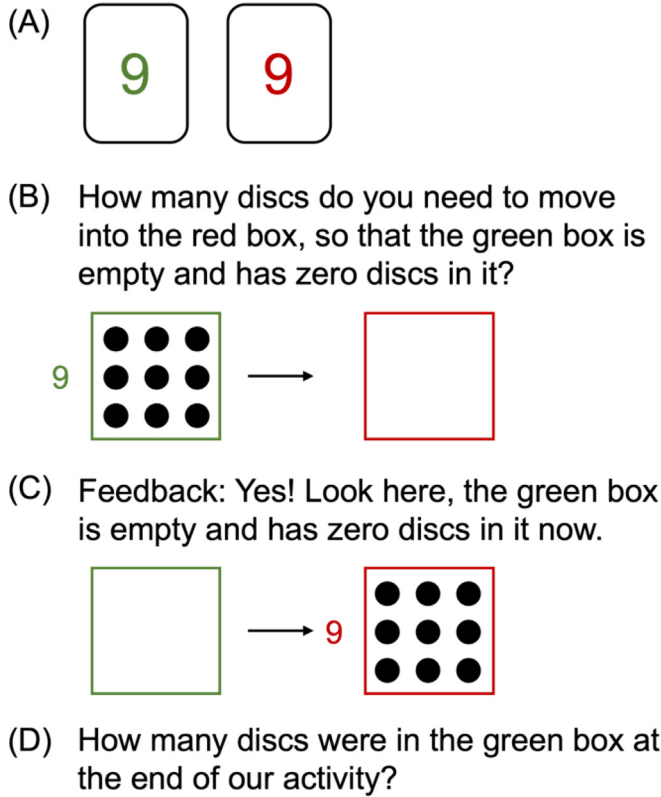


Fig. 3. Sample of card-sorting task used during intervention in the zero-null condition.

integer (-2 vs. 6) or vice versa (-6 vs. 2). These integer sets and pairing rules yielded 44 stimuli (see online [supplementary material](#)). There were four blocks, and each pair appeared once within each block. The blocks varied in whether the stimuli were presented in left-right order (1 vs. 9 , 9 vs. 1).

Comparison predicate was a between-participants factor with two levels: greater and lesser. Participants were randomly assigned to a version of the task that asked them to choose either the greater or lesser number in each pair.

Integer arithmetic fluency

Finally, participants completed an integer arithmetic fluency task on Qualtrics (Tsang et al., 2015). The measure included 26 integer addition problems (e.g., $-4 + 9 = \underline{\quad}$), of which 7 were positive-positive combinations, 16 were positive-negative combinations (including 3 with inverse number pairs, $x + -x$), and 3 were negative-negative combinations (see [Appendix C](#)). Participants were asked to complete as many of the problems as they could in 2 min.

The integer arithmetic fluency task was administered only once, after the posttest. We made this choice because this task was cognitively demanding, and we did not wish to fatigue participants prior to the intervention. Furthermore, we did not have specific hypotheses about intervention effects on arithmetic fluency.

Coding

Null versus symmetry concepts

Three items on the pretest and posttest were coded for zero-null and zero-symmetry concepts. For the two zero questions (What is zero? and Is zero a number?), we coded a zero-null concept if par-

Participants explicitly referred to zero as “nothing” or “no items.” We coded a zero-symmetry concept if participants explicitly referred to zero in the context of positive and negative integers (e.g., “Zero is in the middle of positives and negatives”). For the number line item, we coded a zero-symmetry concept if participants placed 3 at the location that implied that zero was at the center of the number line, and we coded a zero-null concept if participants placed 3 at the location that implied that zero was at the left side of the number line. For analysis purposes, if participants provided a response indicating a symmetry concept on *at least one* of the three items, they were categorized as having a symmetry concept of zero. If participants provided no responses that indicated a symmetry concept and at least one response that indicated a null concept, they were categorized as holding a null concept of zero.

Additive inverse understanding

We coded additive inverse understanding on five items at pretest and posttest. Two items were problems that could be solved using the additive inverse principle. For these items, we coded whether participants reported a strategy that relied on the additive inverse principle (note that there were three items of this type, but we made an error in the materials for one of the posttest items, so we excluded that item and the corresponding pretest item). We considered two types of explanations to reflect strategies based on the additive inverse principle: *recognize inversion* explanations, in which participants noted that a positive number and a negative number with the same absolute value would cancel each other (e.g., “the same positive and negative number cancel each other”), and *zero computation* explanations, in which participants noted that two of the addends in the problem summed to zero (e.g., “a positive number plus the same negative number equals zero”).

Two items were worked examples that had been solved using the additive inverse principle. On these items, participants were asked to explain the step in which the hypothetical solver had used the principle. We coded whether participants mentioned the additive inverse principle in each explanation (e.g., “They found that negative seven would cancel out the seven, and then go down and make that a zero”).

One item asked directly about the additive inverse principle (Why is this problem false: $7 + -8 = 0$?). We coded whether participants referred to the additive inverse principle in their response (e.g., “Because seven minus seven would equal zero, but seven minus eight would equal negative one”).

Integer arithmetic fluency

We evaluated the number of items correctly solved on the symbolic fluency test in the 2 min allotted.

Results

Overview

We first present descriptive analyses of participants' concepts of zero at pretest and how they differ across grade levels. We then examined whether participants' pretest concepts of zero were associated with their performance on tasks assessing knowledge of the additive inverse principle at pretest. Next, we examined effects of the instructional interventions, which involved activities designed to support either a null concept or a symmetry concept of zero. Specifically, we examined whether, relative to the zero-null intervention, the zero-symmetry intervention supported children in (a) developing a symmetry concept of zero and (b) understanding the additive inverse principle. Finally, we considered whether participants' pretest concepts of zero were related to their performance on integer arithmetic and magnitude comparison.

Concepts of zero

We hypothesized that at pretest most participants would have a concept of zero as null, some would have a concept of zero as the symmetry point between positives and negatives, some would display both concepts, and some would display neither concept. As predicted, and supporting Hypoth-

esis 1a, most participants (93%, 67 of 72) expressed a null concept on at least one of the three zero concept items (e.g., by responding that 0 is the smallest number, or by choosing Choice A on the number line item; see [Appendix A](#)). Many participants (56%, 40 of 72) also expressed a symmetry concept on at least one of the three items (e.g., by responding that 0 divides the positive and negative numbers, by choosing Choice B on the number line item). Nearly half (35) of the participants expressed both a null concept and a symmetry concept across the three items. Only 5 participants expressed a symmetry concept without ever expressing a null concept, and every participant expressed one or the other concept at least once across the three items.

We next examined whether participants in higher grades were more likely to express a symmetry concept. We conducted a logistic regression with grade level as the predictor (coded -1 , 0 , and 1 for fifth, sixth, and seventh grade, respectively) and symmetry concept (yes or no) as the outcome. Supporting Hypothesis 1b, the likelihood that participants expressed a symmetry concept on at least one item increased with grade level, $b = 1.31$, $SE = 0.39$, $\chi^2(1, N = 72) = 13.97$, $p < .001$ (see [Fig. 4](#)).

Zero concepts and additive inverse principle knowledge

We next examined whether pretest conceptions of zero were associated with additive inverse principle knowledge at pretest. There were five items in which participants could demonstrate additive inverse understanding, and they received 1 point for each response that demonstrated understanding. We analyzed the data with linear regression, with zero concept category as a predictor and additive inverse score as the dependent variable. As predicted, and supporting Hypothesis 2a, participants who held a symmetry concept of zero had higher additive inverse scores than participants who held a null concept ($M = 2.42$, $SD = 1.38$ vs. $M = 1.53$, $SD = 1.48$), $b = 0.89$, $SE = 0.34$, $F(1, 70) = 7.01$, $p = .010$.

Because there is limited work on measuring knowledge of the additive inverse principle, and to better understand whether certain measures of additive inverse knowledge are more strongly related to concepts of zero, we also considered each of the three additive inverse item types separately. As seen in [Table 1](#) (top row), on the problem-solving items, participants categorized as holding a symmetry concept of zero were significantly more likely than participants categorized as holding a null concept to express a strategy that relied on the additive inverse principle, $b = 0.55$, $SE = 0.19$, $F(1, 70) = 8.14$, $p = .006$. The pattern was similar for the worked examples, but the difference was not significant, $b = 0.32$, $SE = 0.20$, $F(1, 70) = 2.50$, $p = .117$. For the direct question about the principle, relatively few participants ($n = 10$) overall responded by mentioning the additive inverse, and there was little difference in the proportion of participants who did so between participants who held a symmetry concept (15%) and participants who held a null concept (12.5%).

Alternative scoring of problem-solving items

We also considered an alternative approach to scoring use of the additive inverse principle on the problem-solving items, based on participants' reaction times. In accordance with prior literature, we used a 5-s cutoff to categorize problems as having been solved with the additive inverse principle or not ([Fyfe, Matz, Hunt, & Alibali, 2019](#); [Siegler & Stern, 1998](#)). This RT-based categorization relies on the assumption that participants employed the additive inverse principle as a shortcut if they solved the problem quickly because the shortcut involves fewer steps than the more time-intensive strategy of first adding and then subtracting.

We examined the relation between zero concepts and use of the additive inverse principle at pretest, as manifested in RT. As seen in [Table 1](#) (bottom row), on the problem-solving items, participants who held a symmetry concept of zero solved the problems in less than 5 s more often than participants who held a null concept, suggesting that they used the additive inverse principle more often, $b = 0.34$, $SE = 0.17$, $F(1, 70) = 4.43$, $p = .039$.

To check whether RTs were consistent with verbal explanations, we examined whether the two measures were associated. For each of the two problem-solving items, we conducted a logistic regression with RT category (i.e., solved in less than or more than 5 s) as the independent variable and whether participants verbally described the additive inverse principle in their explanation as the dependent variable. RT category and verbal explanations were associated both for the " $6 + 5 + -5$ "

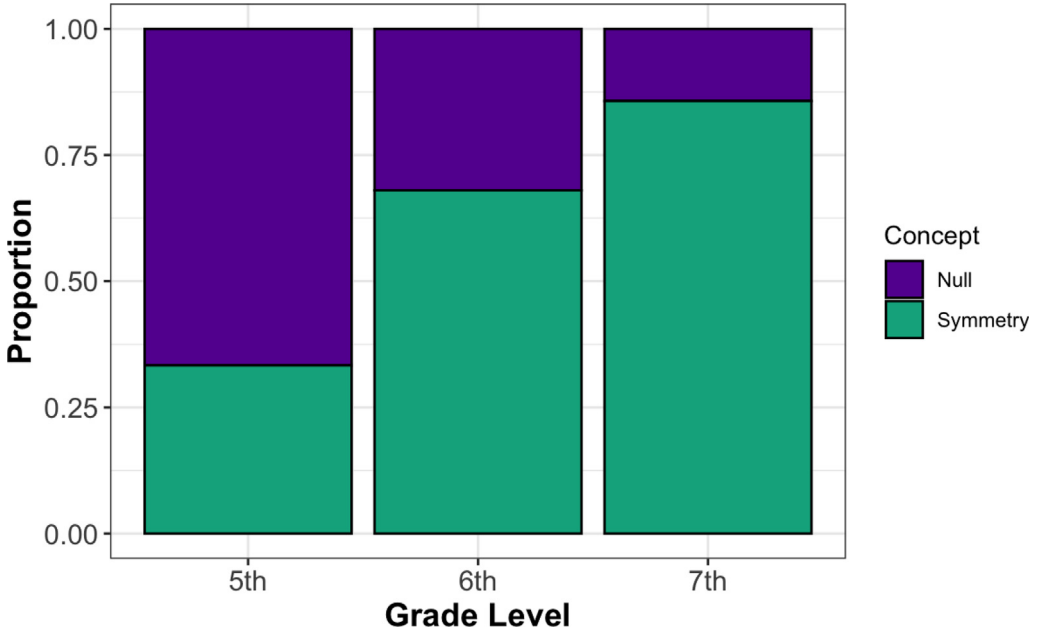


Fig. 4. Proportion of participants at each grade level who displayed symmetry and null concepts of zero at pretest.

Table 1

Demonstration of additive inverse understanding by participants who displayed null and symmetry conceptions of zero on the pretest.

Measure	Overall	Concept of zero		Comparison	P
		Null	Symmetry		
Verbal explanations (out of 2)	0.68 (0.85)	0.38 (0.71)	0.92 (0.89)	$t(70) = 6.74$	.006
Worked example explanations (out of 2)	1.23 (0.85)	1.07 (0.87)	1.35 (0.83)	$t(70) = 1.59$	.117
Concept item (out of 1)	0.14 (0.35)	0.13 (0.34)	0.15 (0.36)	$\chi^2(1) = 0.09$	.760
Reaction time < 5 s (out of 2)	0.54 (0.73)	0.34 (0.55)	0.70 (0.82)	$t(70) = 2.11$	.039

Note. Measures are reported as means (and standard deviations). Participants were classified as displaying a symmetry concept of zero if they expressed symmetry reasoning on one or more of the three zero concept items at pretest.

item, $b = 1.36$, $SE = 0.58$, $\chi^2(1, N = 72) = 5.53$, $p = .019$, and for the “ $27 + 48 + -27$ ” item, $b = 2.60$, $SE = 0.70$, $\chi^2(1, N = 72) = 18.69$, $p < .001$.

Effects of intervention on zero concepts and additive inverse understanding

Concepts of zero

We hypothesized that the intervention emphasizing symmetry would support participants in constructing the symmetry concept of zero (Hypothesis 3a). To test this hypothesis, we fit a generalized linear model with pretest zero concept category (null or symmetry) and condition (zero–null or zero–symmetry) as predictors and posttest zero concept category as the dependent measure. Pretest zero concepts were associated with posttest concepts, $b = 2.47$, $SE = 0.60$, $\chi^2(1, N = 69) = 21.00$, $p < .001$, but the effect of condition on posttest zero concepts was not significant, $b = 0.56$, $SE = 0.59$, $\chi^2(1, N = 69) = 0.92$, $p = .338$. In the zero–null condition, an identical proportion of participants displayed a symmetry concept of zero at pretest and posttest (.60 at each test). In the zero–symmetry condition, a greater proportion of participants displayed a symmetry concept of zero at posttest (.65) than at pretest (.51), but this difference was not significant. Thus, contrary to Hypothesis 3a, a

brief lesson focusing on symmetry did not promote shifts in participants' concepts of zero more than a brief intervention focusing on the null conception (Table 2).

In a set of exploratory analyses, we looked at performance on each of the three items that made up the zero concept assessment separately to examine whether composite scores might be obscuring intervention effects on individual items. As seen in Table 2 (second row), for the "Is zero a number?" item, participants in the zero-null condition were *less* likely to respond affirmatively after the lesson, whereas there was little change among participants in the zero-symmetry condition, $\chi^2(1, N = 72) = 5.84, p = .016$. Also as seen in Table 2, for the number line item, participants in the zero-symmetry condition were *more* likely to respond as if zero were at the center of the number line after the lesson, whereas there was little change on this item among participants in the zero-null condition, $\chi^2(1, N = 72) = 6.54, p = .011$. This finding is fairly unsurprising given that the zero-symmetry lesson used number lines with zero as the midpoint. However, despite these patterns for individual items, pretest-to-posttest changes in overall symmetry scores (out of 3) also did not differ significantly between conditions, $b = 0.29, SE = 0.15, F(1, 69) = 3.55, p = .064$.

Additive inverse principle knowledge

We next examined whether the lesson emphasizing symmetry led to increased understanding of the additive inverse principle at posttest. We conducted a linear regression with pretest additive inverse score and condition as predictors and posttest additive inverse score as the dependent variable. Pretest additive inverse scores were associated with posttest additive inverse scores, $b = 0.70, SE = 0.09, F(1, 69) = 62.92, p < .001$. However, contrary to Hypothesis 3b, posttest additive inverse scores were not higher in the zero-symmetry condition ($M = 2.41, SD = 1.40$) than in the zero-null condition ($M = 2.69, SD = 1.62$), $b = -0.16, SE = 0.26, F(1, 69) = 0.39, p = .536$.

To investigate this null effect further, we considered each of the three item types separately in a linear regression. As seen in Table 3 (top row), participants in the zero-symmetry condition were not significantly more likely than participants in the zero-null condition to express a strategy that relied on the additive inverse principle on the problem-solving items or the worked example items. For the direct question about the principle, there also was little difference between conditions. Thus, despite the symmetry intervention's focus on inverse number pairs and equidistance from zero, there was no evidence that it led to greater gains in additive inverse principle understanding than the intervention focused on the null concept.

Zero concepts and integer arithmetic fluency

We next explored predictors of participants' fluency in solving integer addition problems. One participant was excluded from these analyses because they ended the experiment early and did not complete the integer arithmetic task. To analyze the data, we used a generalized linear mixed-effects model predicting accuracy on each item from pretest zero concepts.¹ On average, participants solved approximately 16 of the 26 problems correctly ($M = 16.24, SD = 6.60, \text{range} = 2\text{--}25$). In support of Hypothesis 2b, pretest zero concepts were associated with performance on the integer arithmetic task, $b = 0.32, SE = 0.09, \chi^2(1, N = 71) = 12.53, p < .001$, with participants who held a symmetry concept solving more problems correctly.

The integer arithmetic test included several different types of problems, including problems that involved solely positive integers. There were 7 positive-positive combinations, 13 positive-negative combinations, 3 additive inverse combinations (e.g., $7 + -7 = __$), and 3 negative-negative combinations. Thus, it was possible that the relations between pretest zero concepts and integer arithmetic performance could be due to positive-positive problems alone. To examine whether this was the case, we examined the association of pretest zero concepts with performance on each of the problem types. Participants who held a symmetry concept of zero at pretest were significantly more accurate than

¹ In a separate analysis, we also included intervention condition (zero-null or zero-symmetry) as a covariate because participants took part in the intervention before they completed the integer arithmetic test, so the intervention plausibly could have affected performance on the integer arithmetic test. There was no significant effect of condition on integer arithmetic fluency, $b = 0.77, SE = 0.51, \chi^2(1, N = 71) = 2.32, p = .128$.

Table 2

Proportions of participants in the zero-null and zero-symmetry conditions who demonstrated a symmetry concept at pretest and posttest.

Item	Zero-Null		Zero-Symmetry		Comparison ^a	p
	Pretest	Posttest	Pretest	Posttest		
“What is zero?”	0.34 (0.48)	0.37 (0.49)	0.38 (0.49)	0.32 (0.47)	$\chi^2(1) = 0.50$	.480
“Is zero a number? Why or why not?”	0.23 (0.43)	0.09 (0.28)	0.22 (0.42)	0.27 (0.45)	$\chi^2(1) = 5.84$	.016
“Place 3 on the number line.”	0.34 (0.48)	0.34 (0.48)	0.14 (0.35)	0.38 (0.49)	$\chi^2(1) = 6.54$	.011

Note. Measures are reported as means (and standard deviations).

^a Effect of condition (zero-null vs. zero-symmetry) on posttest, controlling for pretest.

Table 3

Demonstration of additive inverse understanding at pretest and posttest by participants in the zero-null and zero-symmetry conditions.

Measure	Zero-Null		Zero-Symmetry		Comparison ^a	p
	Pretest	Posttest	Pretest	Posttest		
Verbal explanations (out of 2)	0.69 (0.87)	1.17 (0.95)	0.68 (0.85)	1.35 (0.89)	$t(69) = 1.04$	.300
Worked example explanations (out of 2)	1.31 (0.90)	1.37 (0.88)	1.11 (0.81)	0.92 (0.92)	$t(69) = -1.96$	.054
Concept item (out of 1)	0.11 (0.32)	0.14 (0.36)	0.16 (0.37)	0.14 (0.35)	$\chi^2(1) = 0.08$	.773
Reaction time < 5 s (out of 2)	0.51 (0.70)	1.17 (0.86)	0.57 (0.77)	1.32 (0.82)	$t(69) = 0.71$	.481

Note. Measures are reported as means (and standard deviations).

^a Effect of condition (zero-null vs. zero-symmetry) on posttest, controlling for pretest.

those who held a null concept on all problem types [positive-positive: $b = 0.24$, $SE = 0.08$, $\chi^2(1, N = 71) = 7.06$, $p = .004$; positive-negative: $b = 0.36$, $SE = 0.11$, $\chi^2(1, N = 71) = 11.28$, $p < .001$; additive inverse: $b = 0.41$, $SE = 0.14$, $\chi^2(1, N = 71) = 8.41$, $p = .004$; negative-negative: $b = 0.32$, $SE = 0.14$, $\chi^2(1, N = 71) = 5.06$, $p = .025$].

Zero concepts and integer magnitude comparisons

Finally, we examined performance on the magnitude comparison task. Sixteen participants did not complete this task because they took part in the experiment using a touchscreen device that was incompatible with the software used to display the magnitude comparison task. For consistency with prior work (Varma & Schwartz, 2011), we also excluded 2 participants who performed below chance on the magnitude comparison task (>49% error rate). Thus, analyses are based on data from 54 participants.

Following the analytic procedures used by Varma and Schwartz (2011), we excluded responses outside of the interval of 200 to 2000 ms (6.07% of the data) and responses more than 3 standard deviations from each participant’s mean (0.66% of the data). We also excluded incorrect responses (4.78% of the data). In total, 11.51% of responses were excluded.

Distance effect and concepts of zero

We had two hypotheses regarding distance effects. First, we hypothesized that (4b) participants who held a *null* concept of zero at pretest *would not* show a distance effect on mixed (positive-negative) comparisons because they would view all negative numbers as less than positive numbers, so they would not activate the distinct magnitudes of negative numbers when performing the comparisons. Second, we hypothesized that (4a) participants who held a *symmetry* concept of zero at pretest *would* show an inverse distance effect (far pairs more accurate and faster than near pairs) on mixed comparisons because they would activate magnitudes in order to perform the comparisons (see Varma & Schwartz, 2011). To address these hypotheses, the critical items were the mixed (positive-negative) comparisons. These comparisons were embedded in a full set of comparisons that also included positive, negative, and zero comparisons.

To analyze the data, we fit linear mixed-effects models with reaction time (in milliseconds) as the dependent variable and comparison type (positive, negative, mixed, or zero), distance (near or far), pretest zero concept category (null or symmetry), and all possible interactions as predictors. We

included predicate (greater or lesser) as a covariate.² A full report of the results for all the comparison types is presented in the [supplementary material](#).

For the current aims, the critical prediction was that there would be an interaction of zero concept and distance for mixed comparisons. In contrast, for positive comparisons (the reference category), there should be no interaction of zero concept and distance because all participants should demonstrate the distance effect regardless of zero concept. Thus, evidence to support our prediction would be the three-way interaction of comparison type, distance, and zero concept, and specifically, a data pattern indicating that participants who hold a *null* concept of zero do not show a distance effect on mixed comparisons, but participants who hold a *symmetry* concept of zero do show a distance effect. This interaction was not significant, $F(1, 198) = 0.91, p = .341$. As seen in Fig. 5 (left set of bars), as expected, participants who held a *null* concept of zero did not show a distance effect on mixed comparisons (left panel), in line with Hypothesis 4b. However, contrary to Hypothesis 4a, participants who held a *symmetry* concept of zero also did not show a distance effect on mixed comparisons (Fig. 5, right set of bars). Thus, there was no evidence to support the hypothesis that different zero concepts would be associated with distinct patterns of distance effects in positive–negative magnitude comparisons.

Discussion

This study investigated children's concepts of zero and whether these concepts related to their understanding of integers. Specifically, we asked the following questions. First, what are children's concepts of zero in late elementary and middle school? Second, are children's concepts of zero associated with their understanding of integers, specifically their understanding of the additive inverse principle and their fluency with integer arithmetic? Third, does an intervention emphasizing zero as the symmetry point between positives and negatives enhance children's understanding of zero and of the additive inverse principle, relative to an intervention that focuses on the null conception of zero? Fourth, do children's conceptions of zero predict whether the children demonstrate a distance effect for positive–negative magnitude comparisons?

Empirical summary

We hypothesized that most children would demonstrate a null concept of zero, and some would demonstrate a concept of zero as the symmetry point between the negative and positive integers. Indeed, most participants (93%) demonstrated a null concept on at least one of the three pretest zero concept items, and 56% of participants demonstrated a symmetry concept. Moreover, the proportion of participants who displayed a symmetry concept increased across grade levels. Most children in the United States receive formal instruction on negative integers starting in the sixth grade ([National Governors Association Center for Best Practices & Council of Chief State School, 2010](#)), and our data reveal a shift in zero concepts around sixth grade. Thus, the findings suggest that as children learn about and operate with negatives in school, their concepts of zero shift.

We further hypothesized that children's concepts of zero would be associated with their performance on tasks that tapped understanding of the additive inverse principle and integer arithmetic fluency. Both hypotheses were supported. Participants who displayed a symmetry concept also displayed greater understanding of the additive inverse principle, and they were more successful on integer addition.

We also hypothesized that an intervention focusing on symmetry, and specifically focusing on inverse number pairs, would encourage children to develop a conception of zero as the symmetry point between the positive and negative numbers more so than an intervention focusing on a null conception. However, we found no evidence that our interventions differentially affected participants' zero concepts. There was also no evidence that the symmetry intervention led to changes in understanding of the additive inverse principle.

² In a separate model, we also included intervention condition (zero–null or zero–symmetry) as a covariate. Intervention condition was not a significant predictor of reaction times, $b = 25.50, SE = 43.26, \chi^2(1, N = 54) = 0.35, p = .555$. Thus, there was no evidence that the experimental manipulation affected magnitude comparison performance.

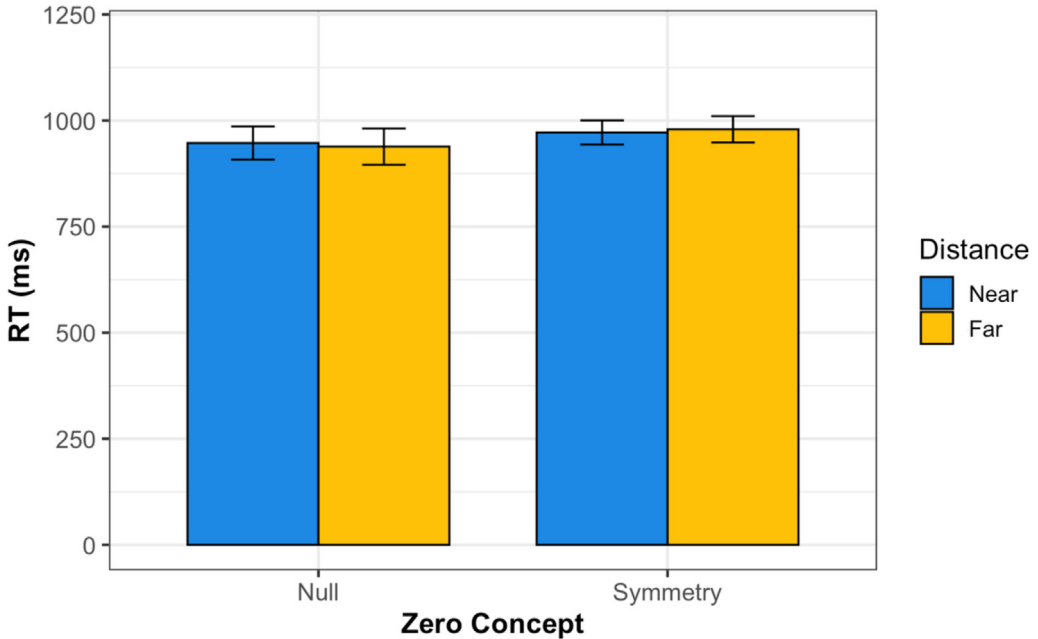


Fig. 5. Reaction times (RTs) for magnitude comparisons that involved a positive integer and a negative integer for participants who provided null and symmetry concepts of zero at pretest. Near-distance stimuli had a distance of 2 or 3, and far-distance stimuli had a distance of 7 or 8.

We had two main hypotheses regarding children's judgments of integer magnitudes in the magnitude comparison task. First, we hypothesized that participants who held a zero-null concept would not show a distance effect on mixed (i.e., negative-positive) comparisons because they would view all negative numbers as less than positive numbers and they would not automatically activate knowledge about the magnitudes of negatives. This hypothesis was supported. Second, we hypothesized that participants who held a zero-symmetry concept *would* show a distance effect on mixed comparisons, specifically an inverse distance effect (Krajcsi & Igács, 2010; Varma & Schwartz, 2011). This hypothesis was not supported; participants with a zero-symmetry concept also did not display any distance effect on mixed comparisons. These findings suggest that, regardless of their zero concepts, children in this age range do not automatically activate the magnitudes of negative integers. These findings are compatible with those of Mock et al. (2019), who reported that sixth-grade students did not automatically associate the negative sign with the left side of space.

Theoretical implications

Our results highlight two distinct conceptions of zero: zero as null and zero as the symmetry point between positive and negative numbers. These two conceptions align with the grounding metaphors described by Lakoff and Núñez (2000). Specifically, the zero as null conception aligns with the metaphor that *numbers are collections of objects*; within this grounding metaphor, zero is the absence of any objects. The zero as the symmetry point between positives and negatives conception aligns with the metaphor that *numbers are locations along a path*; within this metaphor, zero is a point along the path between the positives and negatives—the midpoint of a mental number line that extends to both sides of zero.

Our findings demonstrate that children in the United States tend to progress from the null conception to the symmetry conception in late elementary school or early middle school. The Common Core

standards for Grade 6 require students to describe scenarios with positive and negative quantities and the meaning of zero in each case (e.g., temperature above/below zero; [National Governors Association Center for Best Practices & Council of Chief State School, 2010](#)). Our findings suggest that children's understanding of zero shifts during this time.

Our single brief lesson focused on the symmetry conception did not lead to changes in participants' concepts of zero relative to a lesson focused on the null conception. However, there were some suggestions in participants' performance on individual items that the lessons provoked some changes in participants' thinking. For example, participants in the zero-symmetry condition were more likely than participants in the zero-null condition to respond as if zero were at the center of the number line after the lesson, and participants in the zero-null condition were more likely than participants in the zero-symmetry condition to respond "no" when asked, "Is zero a number?" These findings are compatible with the idea that children's zero concepts are malleable—although changes in such concepts may be gradual. More substantial changes may require more focused instruction or sustained experience with integers.

We found that children's conceptions of zero were associated with their knowledge of related concepts and skills, including the additive inverse principle and their skill at adding integers. These findings align with other work demonstrating associations and causal links between knowledge of concepts and knowledge of problem-solving procedures (see [Rittle-Johnson et al., 2015](#), for a review). Our findings provide further evidence for the associations between concepts and procedures given that participants who held a symmetry concept of zero also demonstrated greater procedural skill in the integer arithmetic task.

We did not find evidence that children in this age range automatically activate magnitude knowledge when comparing positives and negatives given that they did not demonstrate a distance effect for such pairs. This was the case both for children who held a null conception of zero, as was expected, and for children who held a symmetry conception of zero. These findings are consistent with those of [Varma and Schwartz \(2011\)](#), who found that sixth-grade students did not demonstrate a distance effect for positive-negative pairs. In contrast, college-educated adults (who presumably hold a symmetry conception of zero) do show evidence of activating magnitudes in such tasks, as shown both by [Varma and Schwartz \(2011\)](#) and by [Vest & Alibali \(2021\)](#). Taken together, these findings suggest that the development of negative number magnitude understanding is protracted, and it may depend on experience with negative numbers. On its own, a symmetry conception of zero seems not to be enough to provoke automatic activation of negative number magnitudes.

Educational implications

Our findings indicate that understanding of zero shifts in the late elementary and middle school grades. From a practical perspective, recognizing how children's concepts shift with grade level may help teachers to anticipate and address potential misconceptions (such as "zero is not a number") and to foster progress toward more advanced conceptions. It may also be valuable for teachers to know that some students continue to hold *only* a null conception of zero, given that this conception may make learning other related concepts and skills more challenging. For example, if students hold solely a null conception, it may be challenging to learn integer addition and subtraction. More broadly, our findings suggest that understanding of zero (and how zero relates to other integers) should be a target of assessment and instruction during the elementary and middle school years.

It may also be valuable for teachers to be aware of relations between children's zero concepts and their understanding of the additive inverse principle. Our findings indicate that a symmetry concept of zero is closely tied to understanding the additive inverse principle, suggesting that activities that involve the additive inverse principle may in turn support the development of a symmetry conception of zero. By definition, the additive inverse of a given number is a number that, when added to the given number, yields a sum of zero. This definition implies that a number and its additive inverse are equidistant from zero and that therefore their locations on the number line are symmetric around zero. It is possible that learners may benefit from lessons that explicitly delineate these relationships,

for example, by describing the role of zero in identifying a given number's additive inverse and by demonstrating the symmetry around zero of a given number and its inverse using the number line—ideas that could be tested in future intervention studies.

Given the absence of an explicit focus on zero concepts in the Common Core standards for late elementary and middle school ([National Governors Association Center for Best Practices & Council of Chief State School, 2010](#)), it seems likely that there is relatively little instructional attention to zero in this grade band; however, our findings highlight the changing nature of children's zero concepts during this time. We believe that our findings may be valuable for teachers in that they may inform instruction about zero, negative numbers, and the additive inverse principle.

Limitations and future directions

The current findings must be considered in light of the study's limitations. First, our sample was small and homogeneous, consisting of typically developing children who were predominantly White (75%). To enhance generalizability, future research should include larger and more diverse samples. Specifically, researchers should seek to recruit participants with varying mathematical abilities and should include participants from a broader range of socioeconomic, racial, and ethnic backgrounds.

Critically, our inferences about relations between concepts of zero and integer understanding were based on small numbers of items. This was necessary to keep the session length manageable for participants given their ages and given that the experiment was conducted online. Future work should use more comprehensive assessments of children's zero concepts and the additive inverse principle. We are currently developing a more extensive multi-item measure of additive inverse understanding that incorporates both verbal explanations and RTs.

Our study involved a brief online intervention focused on zero concepts, and we did not conduct an a priori power analysis because this was our first attempt to develop an intervention of this sort. We also did not include a true control group. Instead, we sought to develop parallel lessons that focused on distinct concepts of zero (symmetry and null); thus, one condition highlighted the null conception of zero (which most participants were expected to have at the outset of the study), and the other highlighted the symmetry conception. Given that our design involved comparing two distinct interventions, rather than comparing the presence and absence of a single intervention, the inferences we can draw based on condition effects are limited.

The zero–symmetry intervention did not lead to significant gains in participants' zero concepts or the additive inverse principle relative to the zero–null intervention. However, we believe that a future study with a more substantial intervention “dosage” might yield positive effects. Indeed, recent research on integer understanding that involved more intense interventions has demonstrated significant effects. [Tsang and colleagues \(2015\)](#) delivered instruction about integers and symmetry for 4 days, 55 min per day, over a 3-week period, and this intervention led to substantial changes in children's integer concepts. Future research on zero concepts could also implement multiple or longer lessons or could include more extensive number line instruction with positive and negative integers. Future studies could also more explicitly emphasize the concept of symmetry and could use a wider range of tasks or external representations to do so. Recall that [Tsang and colleagues \(2015\)](#) used a specially designed manipulative to highlight the symmetry of the number line. They found that lessons using this manipulative led to greater use of symmetry-based strategies and that use of such strategies was associated with greater accuracy on problems involving negative numbers. Other diagrams, animations, or manipulatives that emphasize symmetry might yield similar effects. We believe that a more intense lesson emphasizing the symmetry conception of zero has the potential to enhance both conceptual and procedural understanding of integers.

Another possible explanation for the lack of intervention effects may be the online context of the study. Past research suggests that in-person lessons are generally more effective than comparable online lessons (e.g., [Menendez, 2022](#)), so it is possible that an intervention like the one we used might yield significant intervention effects in an in-person context. Furthermore, due to the online setting, our intervention was also not highly interactive. Future studies conducted in person could incorporate more hands-on activities into lessons, a perspective we are currently exploring in a follow-up study.

Our study revealed an increase in the proportion of participants who demonstrated a symmetry understanding of zero across the late elementary and middle school grades; however, because we used a cross-sectional design, we could not track the timing of this shift for individual participants, nor could we examine whether understanding of the additive inverse principle “leads” or “follows” the symmetry conception. Future research using longitudinal designs could address these and other related questions about the timing and sequence of the development of understanding of zero and other integer concepts and skills.

Conclusion

To our knowledge, this is the first study to investigate relations between understanding of zero and integer concepts and skills. We found that children educated in the United States tended to progress from a null concept of zero to a symmetry concept during the late elementary and middle school years. Furthermore, children who held a symmetry concept of zero tended to display better understanding of the additive inverse principle as well as greater skill in adding integers. However, we did not find evidence that children’s zero concepts could be enhanced with a single brief online lesson focusing on the symmetry conception relative to a comparable lesson focusing on the null conception. Furthermore, regardless of children’s zero concepts, we did not find evidence that children in this age range automatically activate magnitude knowledge for negative integers. We suggest that future efforts to support the development of integer understanding focus on helping children to make connections among the symmetry conception of zero, the additive inverse principle, and integer magnitudes.

CRedit authorship contribution statement

Nicholas A. Vest: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.
Martha W. Alibali: Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Data availability

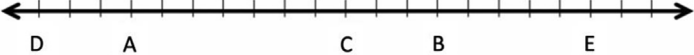
Data will be made available on request.

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Appendix A

Items on pretest and posttest assessments

Item type	Item
Zero concept	What is zero?
Zero concept	Is zero a number? Why or why not?
Additive inverse problems	Solve and explain: $6 + 5 + -5 = \underline{\hspace{1cm}}$ $27 + 48 + -27 = \underline{\hspace{1cm}}$ $31 + -23 + 5 + -5 + -2 + -5 = \underline{\hspace{1cm}}$
Zero concept: Number line	Choose the letter where you think the number 3 is on the number line. 
Additive inverse: Concept	$7 + -8 = 0$ Why is this problem false?
Additive identity: Concept	$3 + 0 = 3$ Why is this problem true?
Worked examples	Step 1: $(4 + 3) + -7 + 8 = \underline{\hspace{1cm}}$ Step 2: $7 + -7 + 8 = \underline{\hspace{1cm}}$ Step 3: $0 + 8 = \underline{\hspace{1cm}}$ Step 4: $8 = \underline{\hspace{1cm}}$ How did the student get from Step 2 to Step 3? Step 1: $23 + -15 + (2 + -10) = \underline{\hspace{1cm}}$ Step 2: $23 + -15 + -8 = \underline{\hspace{1cm}}$ Step 3: $8 + -8 = \underline{\hspace{1cm}}$ Step 4: $0 = \underline{\hspace{1cm}}$ How did the student get from Step 3 to Step 4?

Note. Participants saw isomorphic variations of the additive inverse problems and worked examples for the pretest and posttest.

Appendix B

Fill-in-the-blank statements provided to children during intervention for each condition

Zero–Null condition	Zero–Symmetry condition
A box has 9 discs. <u> </u> discs and then 3 additional discs are pulled out so that there are 0 discs left in the box and it is empty.	(<u> </u> and <u> </u> are both 3 units away from 0.
A box has <u> </u> discs. 4 discs and then 2 additional discs are pulled out so that there are 0 discs left in the box and it is empty.	5 and <u> </u> are both <u> </u> units away from 1.
A box has 4 discs. 2 discs and then <u> </u> additional discs are pulled out so that there are 0 discs left in the box and it is empty.	4 and <u> </u> are both <u> </u> units away from 0.
A box has <u> </u> discs. 5 discs and then 3 additional discs are pulled out so that there are 0 discs left in the box and it is empty.	<u> </u> and <u> </u> are both 4 units away from 3.
A box has 5 discs. 2 discs and then 3 additional discs are pulled out so that there are <u> </u> discs left in the box and it is empty.	8 and -2 are both <u> </u> units away from <u> </u> .

Appendix C

Items on the integer arithmetic fluency task

$9 + 4 =$	$5 + 8 =$	$7 - 4 =$	$11 - 3 =$
$-5 + 2 =$	$1 + -6 =$	$6 + -2 =$	$-1 + 5 =$
$-2 + 4 =$	$4 + -1 =$	$3 + -5 =$	$-9 + 1 =$
$-4 + 4 =$	$-18 + 18 =$	$7 + -7 =$	$0 + -16 =$
$-3 + -6 =$	$-1 + -4 =$	$-5 + -2 =$	$2 - 5 =$
$3 - 4 =$	$1 - 8 =$	$0 + -7 =$	$-5 + 0 =$
$-15 + 13 =$	$-18 + 21 =$		

Appendix D. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jecp.2024.106041>.

References

- Aulet, L. S., & Lourenco, S. F. (2018). The developing mental number line: Does its directionality relate to 5- to 7-year-old children's mathematical abilities? *Frontiers in Psychology*, 9, 1142. <https://doi.org/10.3389/fpsyg.2018.01142>.
- Bialystok, E., & Codd, J. (2000). Representing quantity beyond whole numbers: Some, none, and part. *Canadian Journal of Experimental Psychology/Revue Canadienne de Psychologie Expérimentale*, 54, 117–128. <https://doi.org/10.1037/h0087334>.
- Blair, K. P., Tsang, J. M., & Schwartz, D. L. (2013). The bundling hypothesis: How perception and culture give rise to abstract mathematical concepts. In S. Vosniadou (Ed.), *International handbook of research on conceptual change* (2nd ed., pp. 322–340). Taylor & Francis.
- Bofferding, L. (2014). Negative integer understanding: Characterizing first graders' mental models. *Journal for Research in Mathematics Education*, 45, 194–245. <https://doi.org/10.5951/jresmetheduc.45.2.0194>.
- Dehaene, S., Bossini, S., & Giraux, P. (1993). The mental representation of parity and number magnitude. *Journal of Experimental Psychology: General*, 122, 371–396. <https://doi.org/10.1037/0096-3445.122.3.371>.
- Fischer, M. H., & Shaki, S. (2018). Number concepts: Abstract and embodied. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 373. <https://doi.org/10.1098/rstb.2017.0125> 20170125.
- Fyfe, E. R., Matz, L. E., Hunt, K. M., & Alibali, M. W. (2019). Mathematical thinking in children with developmental language disorder: The roles of pattern skills and verbal working memory. *Journal of Communication Disorders*, 77, 17–30. <https://doi.org/10.1016/j.jcomdis.2018.11.001>.
- Galton, F. (1880). Visualised numerals. *Nature*, 21, 252–256. <https://doi.org/10.1038/021252a0>.
- Gelman, R., & Gallistel, C. R. (1986). *The child's understanding of number*. Harvard University Press.
- Hartmann, J., Herzog, M., & Fritz, A. (2022). Zero—An uncommon number: Preschoolers' conceptual understanding of zero. *International Electronic Journal of Elementary Education*, 14, 353–361. <https://doi.org/10.26822/iejee.2022.249>.
- Krajcsi, A., & Igács, J. (2010). Processing negative numbers by transforming negatives to positive range and by sign shortcut. *European Journal of Cognitive Psychology*, 22, 1021–1038. <https://doi.org/10.1080/09541440903211113>.
- Krajcsi, A., Kojouharova, P., & Lengyel, G. (2021). Development of preschoolers' understanding of zero. *Frontiers in Psychology*, 12, 583734. <https://doi.org/10.3389/fpsyg.2021.583734>.
- Lakoff, G., & Johnson, M. (1980). The metaphorical structure of the human conceptual system. *Cognitive Science*, 4(2), 195–208. https://doi.org/10.1207/s15516709cog0402_4.
- Lakoff, G., & Núñez, R. (2000). *Where mathematics comes from*. Basic Books.
- Lappan, G., & Wheeler, M. M. (1987). Children's understanding of zero and infinity. *The Arithmetic Teacher*, 35, 42–44. <https://doi.org/10.5951/AT.35.3.0042>.
- Menendez, D. (2022). *Not only what you see but where and when you see it: Factors influencing learning with visualizations* [Doctoral dissertation, University of Wisconsin-Madison]. Proquest.
- Mock, J., Huber, S., Cress, U., Nuerk, H.-C., & Moeller, K. (2019). Negative numbers are not yet automatically associated with space in 6th graders. *Journal of Cognition and Development*, 20, 611–633. <https://doi.org/10.1080/15248372.2019.1639714>.
- Moyer, R. S., & Landauer, T. K. (1967). Time required for judgements of numerical inequality. *Nature*, 215, 1519–1520. <https://doi.org/10.1038/2151519a0>.
- National Governors Association Center for Best Practices & Council of Chief State School Officers. (2010). *Common Core State Standards for mathematics*. Retrieved from <https://learning.illustrativemathematics.org/wp-content/uploads/2022/11/ADA-Compliant-Math-Standards.pdf>
- Newcombe, N. S., Levine, S. C., & Mix, K. S. (2015). Thinking about quantity: The intertwined development of spatial and numerical cognition. *Wiley Interdisciplinary Reviews: Cognitive Science*, 6, 491–505. <https://doi.org/10.1002/wcs.1369>.
- Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., Kastman, E., & Lindeløv, J. K. (2019). PsychoPy2: Experiments in behavior made easy. *Behavior Research Methods*, 51, 195–203. <https://doi.org/10.3758/s13428-018-01193-y>.
- Piaget, J. (1965). *The child's conception of number*. Norton.
- Rittle-Johnson, B., Schneider, M., & Star, J. R. (2015). Not a one-way street: Bidirectional relations between procedural and conceptual knowledge of mathematics. *Educational Psychology Review*, 27, 587–597. <https://doi.org/10.1007/s10648-015-9302-x>.
- Sarnecka, B. W., & Carey, S. (2008). How counting represents number: What children must learn and when they learn it. *Cognition*, 108, 662–674. <https://doi.org/10.1016/j.cognition.2008.05.007>.
- Sella, F., Sader, E., Lollot, S., & Cohen Kadosh, R. (2016). Basic and advanced numerical performances relate to mathematical expertise but are fully mediated by visuospatial skills. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 42, 1458–1472. <https://doi.org/10.1037/xlm0000249>.
- Siegler, R. S., & Stern, E. (1998). Conscious and unconscious strategy discoveries: A microgenetic analysis. *Journal of Experimental Psychology: General*, 127, 377–397. <https://doi.org/10.1037/0096-3445.127.4.377>.
- Tam, Y. P., Wong, T.-T.-Y., & Chan, W. W. L. (2019). The relation between spatial skills and mathematical abilities: The mediating role of mental number line representation. *Contemporary Educational Psychology*, 56, 14–24. <https://doi.org/10.1016/j.cedpsych.2018.10.007>.
- Tsang, J. M., Blair, K. P., Bofferding, L., & Schwartz, D. L. (2015). Learning to “see” less than nothing: Putting perceptual skills to work for learning numerical structure. *Cognition and Instruction*, 33, 154–197. <https://doi.org/10.1080/07370008.2015.1038539>.
- Varma, S., Blair, K. P., & Schwartz, D. L. (2019). Cognitive science foundations of integer understanding and instruction. In A. Norton & M. W. Alibali (Eds.), *Constructing number* (pp. 307–327). Springer. https://doi.org/10.1007/978-3-030-00491-0_14.
- Varma, S., & Schwartz, D. L. (2011). The mental representation of integers: An abstract-to-concrete shift in the understanding of mathematical concepts. *Cognition*, 121, 363–385. <https://doi.org/10.1016/j.cognition.2011.08.005>.
- Vest, N. A., & Alibali, M. W. (2021). *The mental representation of integers: Further evidence for the negative number line as a reflection of the natural number line*. In Proceedings of the Annual Meeting of the Cognitive Science Society (Vol. 43, No. 43).
- Wellman, H. M., & Miller, K. F. (1986). Thinking about nothing: Development of concepts of zero. *British Journal of Developmental Psychology*, 4, 31–42. <https://doi.org/10.1111/j.2044-835X.1986.tb00995.x>.

- Wheeler, M. M., & Feghali, I. (1983). Much ado about nothing: Preservice elementary school teachers' concept of zero. *Journal for Research in Mathematics Education*, 14(3), 147–155. <https://doi.org/10.2307/748378>.
- Wood, G., Willmes, K., Nuerk, H. C., & Fischer, M. H. (2008). On the cognitive link between space and number: A meta-analysis of the SNARC effect. *Psychology Science Quarterly*, 50, 489–525.
- Wynn, K. (1998). Psychological foundations of number: Numerical competence in human infants. *Trends in Cognitive Sciences*, 2, 296–303. [https://doi.org/10.1016/S1364-6613\(98\)01203-0](https://doi.org/10.1016/S1364-6613(98)01203-0).
- Young, L. K., & Booth, J. L. (2015). Student magnitude knowledge of negative numbers. *Journal of Numerical Cognition*, 1, 38–55. <https://doi.org/10.5964/jnc.v1i1.7>.